E-commerce Revenue & Delivery Analytics

End-to-End Project Documentation

Tools: SQL • Python / Pandas • Power BI • DAX

Purpose
This project uses the Olist Brazilian e-commerce dataset to analyze item revenue, order trends, product categories, state-level performance, delivery delays, and payment-reconciliation categories through an interactive Power BI dashboard.

Business focus
Investigate why strong sales activity may not automatically translate into better profitability. The available data supports analysis of revenue and operational risk indicators; it does not contain complete product costs and operating expenses to calculate actual profit.

1. Project Overview & Business Problem

E-commerce Revenue & Delivery Analytics is an end-to-end analytics project built around a business question: “Why can an e-commerce business generate sales but still struggle to improve profitability?” Revenue alone does not establish profitability. Shipping costs, delivery delays, product mix, and payment exceptions may affect business performance and should be investigated with evidence.

Business Questions

- How do item revenue and order volume change over time?
- Which product categories and states contribute the most item revenue?
- Which states show higher late-delivery rates?
- How are records distributed across payment-reconciliation categories?
- Where should the business investigate freight costs, delivery performance, and data exceptions?

Project Objectives

- Prepare and relate the required Olist tables for analysis.
- Build reusable Power BI measures and calculated fields for key KPIs.
- Create an interactive dashboard with Year, Month, and State slicers.
- Compare revenue trends, category performance, state revenue, delivery rates, and reconciliation categories.
- Translate observed patterns into practical recommendations while clearly documenting data limitations.

Technology Stack

Tool	Role in the project
SQL	Data exploration, aggregation, joins, and validation queries.
Python / Pandas	Data profiling, cleaning checks, derived metrics, and exploratory analysis.
Power BI	Data model, KPI cards, interactive visuals, and dashboard delivery.
DAX	Measures and calculated columns for revenue, orders, delivery rates, and regional grouping.
Olist CSV datasets	Source data for customers, orders, order items, payments, products, and related attributes.

2. Data Sources & Extraction

The project uses public Olist Brazilian e-commerce CSV datasets. The Power BI model combines customer, order, order-item/revenue, delivery, and payment-related information. Table names in the working model include customers_dataset, orders_dataset, order_analysis, monthly_business, monthly_delivery, category_performance, state_delivery_summary, customer_purchase_counts, and project_kpi_summary.

Source / model table	Main analytical use
customers_dataset	Customer identifiers, city, and customer_state.
orders_dataset	Order identifiers, customer links, order status, and order/delivery timestamps.
order_analysis	Order-level analysis fields, including item_revenue, freight/payment fields, and reconciliation_status.
monthly_business	Monthly revenue and order trend metrics.
monthly_delivery	Monthly late-delivery metrics.
category_performance	Category-level item revenue, freight value, item count, and revenue share.
state_delivery_summary	State-level late delivery percentage, late orders, and orders analyzed.
DimMonth / project_kpi_summary	Date attributes and summarized KPI values used in the report.

Extraction and Profiling Approach

- Import the relevant CSV files into Power BI and inspect column names and data types.
- Check row counts, distinct keys, null values, duplicates, date ranges, and numeric fields.

- Relate customers_dataset to orders_dataset using customer_id.
- Relate orders_dataset to order_analysis using order_id where the model's key uniqueness and cardinality support the relationship.
- Keep source data and derived summary tables distinguishable in the model and document the definition of each KPI.

Model validation note
Before publishing, confirm that order_id and customer_id relationships have the expected cardinality and filter direction. Also reconcile the Total Revenue KPI against the revenue total shown in the Revenue by Region visual; the working dashboard has shown approximately 13.54M and 13.59M respectively, so the difference should be investigated rather than silently treated as a match.

3. Data Cleaning & Transformation

Data preparation is required to make the source tables suitable for analysis. Cleaning decisions should preserve the source values and avoid changing business meaning without an explicit rule.

Checks and transformations

- Confirm appropriate data types for identifiers, timestamps, amounts, counts, and percentages.
- Inspect missing customer/order identifiers and review duplicate records before removing any rows.
- Standardize text fields where needed for consistent category and state labels.
- Validate revenue and freight fields for unexpected zero, negative, or unusually high values.
- Create monthly summary fields and category-level summaries for dashboard visuals.
- Map Brazilian state abbreviations to the five geographic regions for the Revenue by Region visual.
- Use a consistent scale for percentage fields. Values stored as 12.6 percentage points must not be formatted directly as a fraction-based percentage, or they may display as 1260%.

Example transformation flow

Workflow
Source CSVs → Profile data → Validate keys and values → Transform / aggregate → Create relationships and measures → Build visuals → Reconcile results

Data-quality considerations

A high freight-to-item-revenue ratio can flag an order for investigation, but it does not alone prove a loss. Returns, discounts, product costs, and operating expenses should be considered before making a profitability conclusion.

4. SQL Analysis & Validation

SQL can be used to independently verify Power BI results and answer business questions before visualizing them. The examples below use generic table/column names; adapt them to the exact names in the local Olist model.

Monthly item revenue

```
SELECT
    YEAR(order_date) AS order_year,
    MONTH(order_date) AS order_month,
    SUM(item_revenue) AS total_item_revenue,
    COUNT(DISTINCT order_id) AS total_orders
FROM order_analysis
GROUP BY YEAR(order_date), MONTH(order_date)
ORDER BY order_year, order_month;
```

Top states by revenue

```
SELECT TOP 5
    customer_state,
    SUM(item_revenue) AS total_item_revenue
FROM order_analysis
GROUP BY customer_state
ORDER BY total_item_revenue DESC;
```

Validation checklist

- Compare source and report row counts.
- Check distinct order counts and duplicate order keys.
- Recalculate monthly item revenue and compare it with monthly_business.
- Compare category and state totals with their summary tables.
- Validate delivery-rate numerator and denominator definitions.
- Reconcile the Total Revenue KPI and donut-chart revenue total before publishing.

5. Power BI Data Model & DAX

The report uses related customer and order data plus analytical summary tables. Relationships should be simple, active where required, and validated for the intended filter flow. Avoid ambiguous many-to-many relationships unless the data model requires them and the behavior is tested.

Key measures used in the report

Measure / KPI	Definition / purpose
Total Revenue	Sum of item_revenue used for the revenue KPI; confirm the exact source table and aggregation.
Total Orders	Order count shown in the dashboard; confirm whether the measure uses distinct order IDs or a validated summary.
Late Delivery Rate	Late orders divided by orders analyzed, using the agreed definition and correct percentage scale.
Average Delivery Days	Average elapsed time from order to delivery for eligible delivered orders.
Total Item Revenue	SUM of order_analysis[item_revenue], used for state-level revenue analysis.

Example DAX

```
Total Item Revenue = SUM(order_analysis[item_revenue])
```

```
Late Delivery Rate % = DIVIDE([Late Orders], [Orders Analyzed], 0)
```

The second expression is illustrative: the measure names and underlying aggregation must match the actual model. If a source column stores percentage points such as 12.6, use a consistent numeric scale and do not apply percentage formatting as if the stored value were 0.126.

6. Power BI Dashboard Design

The completed report is a single-page executive dashboard with a dark navy theme, blue/orange chart accents, four KPI cards, and Year, Month, and State slicers. The slicers were tested and observed to filter the report visuals.

Dashboard element	Purpose
Title banner	E-commerce Sales & Delivery Dashboard identity.
Year / Month / State slicers	Allow users to focus the report on selected periods and states.
Total Revenue KPI	Shows item-revenue total in the current filter context.
Total Orders KPI	Shows the order-volume KPI.
Late Delivery Rate KPI	Highlights delivery performance.
Average Delivery Days KPI	Shows average delivery time.
Monthly Revenue Trend	Shows monthly item revenue over time.
Revenue by Product Category	Compares item revenue across categories.
Revenue by Region	Shows the distribution of item revenue across mapped Brazilian regions.

Top 5 States by Revenue	Ranks leading states by item revenue.
Payment Reconciliation	Shows record counts by reconciliation_status category.
Late Delivery by State	Ranks states by late-delivery percentage.

Design principles applied

- Use a consistent dark background and readable high-contrast labels.
- Keep KPI cards aligned and place visuals in a clear reading order.
- Use blue for revenue-focused charts and orange for delivery-risk emphasis.
- Limit category and state ranking charts to the intended number of items.
- Test slicers to confirm they affect the expected visuals.

7. Business Findings & Recommendations

The dashboard supports investigation of revenue trends, category mix, state performance, delivery delays, and reconciliation exceptions. Findings below should be read as evidence from the current analysis, not as proof of realized profit or loss.

Observed dashboard findings

- The dashboard displays approximately R$13.54M in the Total Revenue KPI and approximately 99K total orders. The revenue KPI and the region visual have shown different totals in the working report, so final revenue figures require reconciliation.
- The overall Late Delivery Rate KPI displays 8.11%, and Average Delivery Days displays 12.50 days.
- State-level delivery rates vary. The current Late Delivery by State visual shows AL, MA, PI, CE, and SE among the highest displayed rates; verify the aggregation and source values before external publication.
- The Payment Reconciliation visual is dominated by “Items and payments.” The current counts discussed during development are 98,665 items-and-payments records, 775 payments-only records, and 1 items-only record; verify the category counts and total against the final model before publishing.
- The dashboard ranks states and categories by item revenue, enabling further investigation of geographic and product concentration.

Recommended business actions

Finding / risk indicator	Recommended action	How to measure impact
High freight relative to item revenue	Investigate outlier orders, shipping lanes, product dimensions, and carrier rates.	Freight per order; freight-to-item-revenue ratio; margin after freight where cost data is available.
Higher late-delivery states	Review carrier performance, dispatch timing, transit distance, and regional capacity.	Late-delivery rate, average delivery days, and on-time delivery by state.
Revenue per order changes	Review product mix, discounts, bundles, and order composition.	Average item revenue per order and contribution margin after costs.
Payment reconciliation exceptions	Audit order status, payment records, cancellations, and unmatched records.	Exception count, exception value, and resolution time.
Revenue concentrated in certain states/categories	Compare customer demand, marketing spend, delivery costs, and product availability.	Revenue growth and contribution margin by state/category.

Profitability limitation
The available project data does not include complete product purchase costs and all operating expenses. Therefore, this project does not calculate verified gross profit, net profit, or profit margin and does not claim that profit has already increased. It identifies operational indicators and follow-up analyses that may help the business improve profitability.

8. Validation, Performance & Deployment

Validation checklist

- Confirm row counts and distinct keys for the source tables.
- Confirm customer_id and order_id relationships and their cardinality.
- Reconcile revenue KPI against the revenue visual and source summary.
- Validate percentage storage scale and formatting for late-delivery measures.
- Check Top 5 state rankings and category rankings against the measure used for sorting.
- Test Year, Month, and State slicers across all intended visuals.
- Check labels, chart titles, sorting, and visual readability at report zoom levels.

Performance and access considerations

- Remove unused columns from the model where they are not required.
- Prefer reusable measures for dynamic calculations when appropriate.
- Avoid unnecessary calculated columns and complex relationships.
- Protect customer identifiers and avoid publishing sensitive row-level data.
- If publishing to Power BI Service, configure workspace access and refresh behavior according to the source and account setup.

9. Challenges, Deliverables & Interview Summary

Challenge	Approach
Different source-table grains	Use key relationships and distinguish order-level tables from monthly/state/category summaries.
Revenue totals do not match across visuals	Trace source table, aggregation, relationships, and filters; publish only after reconciliation.
Percentage displays incorrectly	Check whether values are stored as fractions or percentage points before formatting.
State codes are abbreviated	Retain source codes or add a clearly documented display mapping.
Profitability cannot be directly measured	Document missing cost/expense data and frame recommendations as hypotheses to validate.

Final deliverables

- Power BI dashboard with KPI cards, slicers, and six core visuals.
- Project documentation and KPI definitions.
- SQL validation queries and analysis examples.
- Python/Pandas data profiling and analysis workflow (where used in the project).
- Documented business findings, recommendations, and data limitations.
- GitHub repository and LinkedIn project showcase (planned publication steps).

End-to-end project flow

Project flow
Business question → Source data profiling → Data cleaning and transformation → SQL/Python analysis → Power BI relationships and measures → Interactive dashboard → Validation → Business findings → Recommendations